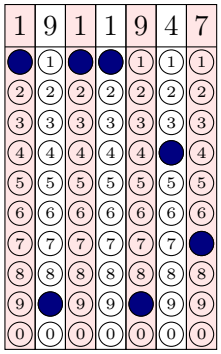
TEST VARIANTI RAQAMI: **1911947**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



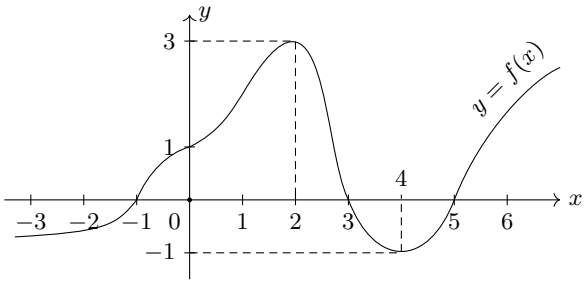
1. $f(x) = (x^2 + x) \cdot \sqrt{x^2 + 1}$ funksiyaning $x_0 = 0$ nuqtadagi hosilasini toping.

A) -1 B) 0 C) 1 D) 2

2. $y = 3x^2 - 6x + 7$ kvadrat funksiyaning abssissa o‘qiga nisbatan simmetrik funksiyasini aniqlang.

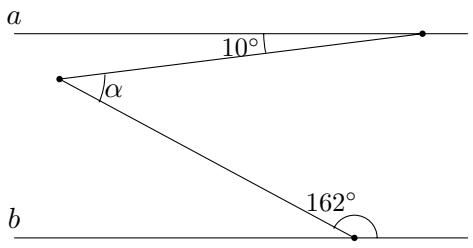
A) $y = -3x^2 + 6x - 7$ B) $y = 3x^2 + 6x + 7$
C) $y = 3x^2 - 6x + 7$ D) $y = -3x^2 - 6x - 7$

3. Funksiya grafigidan foydalanib, $(-3; 6)$ oraliqda $f(x) \cdot f'(x) \le 0$ tengsizlikning eng katta manfiy butun yechimini toping.



A) *manfiy yechimga ega emas* B) -1
C) -3 D) -2

4. a va b parallel to‘g‘ri chiziqlar. Rasmdan foydalanib α burchakning qiymatini toping.



A) *aniqlab bo‘lmaydi* B) 18° C) 24°
D) 28°

5. $\int \dfrac{1}{\sin^2(2x-3)}dx$ integralni hisoblang.

A) $\dfrac{ctg\,(3-2x)}{3} + C$ B) $-\dfrac{ctg\,(3-2x)}{3} + C$
C) $-\dfrac{ctg\,(3-2x)}{2} + C$ D) $\dfrac{ctg\,(3-2x)}{2} + C$

6. Agar $f(x) = \log_2 x^3 + 1$ bo‘lsa, $f(2) + f\left(\dfrac{1}{x}\right) = f(x)$ tenglamani yeching.

A) $\sqrt[3]{4}$ B) $\sqrt[4]{8}$ C) $2\sqrt{2}$ D) 2

7. Radiusi 3 va 4 ga teng bo‘lgan ikki shar markazlari orasidagi masofa 5 ga teng. Shar sirtlari kesishishidan hosil bo‘lgan aylananing uzunligini toping.

A) 4π B) $4,8\pi$ C) $2,4\pi$ D) 5π

8. $\sqrt{(3x-2)^2} + \sqrt[3]{(3x-2)^3} + \sqrt[4]{(3x-2)^4} \ge 4$ tengsizlikning eng katta manfiy butun yechimi bilan eng kichik musbat butun yechimi yig‘indisini toping.

A) *3* B) *1* C) *tengsizlik yechimga ega emas*
D) *4*

9. $(7; -12)$ nuqtaning ordinatalar o‘qiga nisbatan simmetrik bo‘lgan nuqtasini toping.

A) *(7; 12)* B) *(12; −7)* C) *(−7; 12)*
D) *(−7; −12)*

10. Teng yonli uchburchakning ikki tomoni 2 va 5 ga teng. Uchburchakning eng kichik medianasi uzunligini toping.

A) $\dfrac{\sqrt{33}}{2}$ B) $\sqrt{33}$ C) *6,5* D) *13*

11. $\cos^2 2x = \dfrac{2+\sqrt{3}}{4}$ tenglamaning eng kichik musbat yechimini toping.

A) $\dfrac{\pi}{3}$ B) $\dfrac{2\pi}{3}$ C) $\dfrac{\pi}{12}$ D) $\dfrac{\pi}{24}$

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12. Umumiy hadi $x_n = 3 + 5n - n^2$ formula bo‘yicha berilgan sonli ketma-ketlikning eng katta hadi 15 dan qanchaga kam?

A) *6* B) *5,75* C) *9,25* D) *13*

13. Hisoblang:

$$\underbrace{2,2+2,2+...+2,2}_{8\ ta}+\underbrace{1,1+1,1+...+1,1}_{16\ ta}$$

A) *17,6* B) *52,8* C) *70,4* D) *35,2*

14. 3 dm×4 dm×8 dm o‘lchamli to‘g‘ri burchakli parallelepipedlardan nechtasini terib eng kichik hajmli kub hosil qilish mumkin?

A) *180* B) *144* C) *216* D) *72*

15. 0; 1; 2; 3; 4; 5 raqamlardan jami nechta raqamlari takrorlanmaydigan 3 xonali sonlar tuzish mumkin?

A) *216* B) *180* C) *100* D) *125*

16. Agar $a = 6^{3x-2y}$ va $b = 6^{3x+2y}$ bo‘lsa, $4 \cdot 6^x + 3 \cdot 6^y$ ni a va b orqali ifodalang.

A) $4 \cdot \sqrt[4]{ab} + 3 \cdot \sqrt[6]{\dfrac{a}{b}}$ B) $4 \cdot \sqrt[4]{ab} + 3 \cdot \sqrt[3]{\dfrac{a}{b}}$

C) $4 \cdot \sqrt[3]{ab} + 3 \cdot \sqrt{\dfrac{b}{a}}$ D) $4 \cdot \sqrt[6]{ab} + 3 \cdot \sqrt[4]{\dfrac{b}{a}}$

17. $A = \{a;\ b;\ c;\ d;\ e;\ f\}$ to‘plamning elementlari soni 2 ta bo‘lgan nechta qism to‘plami bor?

A) *16* B) *8* C) *4* D) *15*

18. $\left(\dfrac{m^{\frac{3}{4}} - m^{\frac{1}{4}}}{1 - m^{\frac{1}{2}}} + \dfrac{1 + m^{\frac{1}{2}}}{m^{\frac{1}{4}}}\right)^2 :$
$: \left(1 + 2m^{-\frac{1}{2}} + m^{-1}\right)^{0,5}$ ifodaning $m = 36^{-1}$ bo‘lgandagi qiymatini toping.

A) $\dfrac{1}{7}$ B) 5 C) $\dfrac{1}{6}$ D) $\dfrac{6}{7}$

19. $\left(x^2 + 5x + 1\right)^2 + 2x^2 + 10x = 1$ tenglama nechta haqiqiy ildizga ega?

A) *1* B) *2* C) *4* D) *3*

20. $\sqrt[3]{y^2} = \sqrt[3]{y} + 12$ tenglamaning kichik ildizi y_0 bo‘lsa, $y_0 + 14$ ning qiymatini toping.

A) *−13* B) *22* C) *15* D) *13*

21. Kitobning narxi 12500 so‘m edi. Dastlab uning narxi 16% ga arzonlashdi, so‘ng 1850 so‘mga qimmatlashdi. Kitobning oxirgi narxi necha so‘m bo‘ldi?

A) *12350* B) *12450* C) *12260* D) *12220*

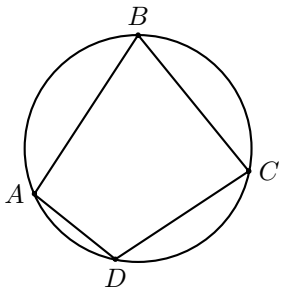
22. Tenglamani yeching:

$(\sin(x + 1,5\pi) - 1) \cdot \operatorname{tg}\dfrac{x}{2} = 0.$

A) $2\pi k,\ k \in Z$ B) $\pi + 2\pi k,\ k \in Z$

C) $\pi k,\ k \in Z$ D) $\dfrac{\pi}{2} + \pi k,\ k \in Z$

23. Rasmda $ABCD$ to‘rtburchak va unga tashqi chizilgan aylana tasvirlangan. Agar AB=5, BC=4, CD=3 va AD=2 bo‘lsa, $\angle ABC$ ning kosinusini toping.



A) $\dfrac{7}{13}$ B) $\dfrac{8}{13}$ C) $\dfrac{5}{13}$ D) $\dfrac{6}{13}$

24. $f(x) = (x + 1)^2 + 4$ parabola uchining koordinatalarini toping.

A) $(-1; 4)$ B) $(1; 4)$ C) $(1; -4)$
D) $(-1; -4)$

25. Agar a va b ratsional sonlar uchun $a + b \cdot \dfrac{\sqrt{3}}{3} = 3$ bo‘lsa, u holda $a^2 + b^2$ ifodaning qiymatini toping.

A) *7* B) *27* C) *9* D) *13*

26. Mis, rux va qo‘rg‘oshindan iborat bo‘lgan 15 kg li qotishmaning tarkibida 20% mis bo‘lib, rux va qo‘rg‘oshinning og‘irliklari mos ravishda 5:1 nisbatda. Qotishmadagi ruxning og‘irligi misning og‘irligidan necha kilogrammga ko‘p?

A) *4,5* B) *7* C) *6* D) *5*

27. Ikki natural sonning EKUKi va EKUBi mos ravishda 420 va 35 ga teng bo‘lsa, ularning ko‘paytmasi 35^2 dan necha marta katta?

A) *16* B) *15* C) *10* D) *12*

28. Hisoblang: $\dfrac{\sqrt{32} + \sqrt{50} - \sqrt{72}}{\sqrt{18}}$

A) 2 B) 1 C) $\dfrac{\sqrt{5}}{3}$ D) 3

29. $\left(\dfrac{2}{3\sqrt{3} - 1}\right)^{x^2 - (2x+1)^2} \le 1$ tengsizlikni yeching.

A) $\left[-1; -\dfrac{2}{3}\right]$ B) $(-\infty; -1] \cup \left[-\dfrac{2}{3}; \infty\right)$

C) $(-\infty; -1] \cup \left[-\dfrac{1}{3}; \infty\right)$ D) $\left[-1; -\dfrac{1}{3}\right]$

30. $\left(-1,5a^2b^3\right)^2 \cdot \left(-100a^3b^4\right)$ ifodani soddalashtiring.

A) $-225a^7b^{10}$ B) $225a^7b^9$ C) $-225a^7b^9$
D) $225a^7b^{10}$

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Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 77-bet 7	7 – 110-bet 31	13 – 12-bet 25	19 – 64-bet 48	25 – 14-bet 43
2 – 75-bet 58	8 – 69-bet 27	14 – 110-bet 33	20 – 60-bet 13	26 – 21-bet 44
3 – 81-bet 38	9 – 116-bet 15	15 – 122-bet 1	21 – 19-bet 26	27 – 9-bet 26
4 – 93-bet 22	10 – 99-bet 71	16 – 26-bet 41	22 – 44-bet 96	28 – 23-bet 11
5 – 85-bet 3	11 – 36-bet 24	17 – 119-bet 3	23 – 104-bet 103	29 – 53-bet 8
6 – 56-bet 7	12 – 34-bet 40	18 – 50-bet 54	24 – 72-bet 33	30 – 45-bet 8

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